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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

GPA-CGPA Calculator

Submitted by

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Under the Guidance of

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Academic Year: **2025 – 2026**

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Semester: **I**

DECLARATION

I hereby declare that the project entitled **"GPA-CGPA Calculator"** submitted in partial fulfillment of the requirements for the course **JAVA PROGRAMMING** is a record of my original work carried out under the guidance of **DHILEEPAN A**, Technical Trainer, Technical Department, KG College of Arts and Science.

This project has not been submitted previously for any degree or diploma.

April **2026**

Coimbatore

ARJUN SANTHOSH

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April 2026

Coimbatore

_____

(Your Name)

CERTIFICATE

This is to certify that the project entitled **"GPA-CGPA Calculator"** is a bona fide work carried out by **ARJUN SANTHOSH**, I YEAR **B.Sc IT**, KG College of Arts & Science, in partial fulfillment of JAVA PROGRAMMING course requirements

Place: Coimbator

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

Head of the Department
(Technical Head, KGCAS)

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ABSTRACT

The CGPA and GPA Calculator is a Java-based console application developed to help students calculate and track their academic performance efficiently.

Many students find it difficult to calculate their GPA (Grade Point Average) and CGPA (Cumulative Grade Point Average) manually. This system helps students calculate their semester GPA and overall CGPA by entering subject marks and credit details.

The system allows students to:

- **Calculate GPA for a semester.**
- **Calculate overall CGPA.**
- **Enter subject marks and credits.**
- **View subject-wise grade points.**
- **Display GPA and CGPA results.**

The system is developed using Core Java concepts such as:

- **Classes and Objects**
- **ArrayList**
- **Loops**
- **Conditional Statements**
- **Methods**
- **User Input using Scanner**
-

INTRODUCTION

In today's competitive world, students apply for many internships and job opportunities. Managing these applications manually becomes difficult and confusing.

The gpa-cgpa calculator helps students maintain records of their applications in a simple and structured way.

This project is developed using Java programming language and works as a menu-driven console application.

It improves organization and helps students track:

- **applied companies**
- **job roles**
- **interview status**
- **selection status**

1.1 OBJECTIVE

Objectives of this project:

- **To develop a job tracking system using Java**
- **To store cgpa-gpa calculator details**
- **To track application status**
- **To update job status easily**
- **To display all application details**
- **To demonstrate usage of ArrayList and classes**

1.2 OVERVIEW

This system allows students to:

- **Enter company name**
- **Enter job role**
- **Enter application status**
- **View stored applications**
- **Update job status**

The system stores application data temporarily using ArrayList.

It is simple, fast, and easy to use.

1.3 FEATURES

Main features include:

Enter subject marks and credits

- **Calculate semester GPA**
- **Calculate overall CGPA**
- **View subject-wise grade details**
- **Menu-driven interface**

PROGRAM ANALYSIS

Program analysis helps understand the problem and provides a computerized solution.

Students face problems like:

- **Difficulty in calculating GPA manually**
- **Confusion in calculating overall CGPA**
- **Difficulty in tracking semester-wise academic performance**

This system solves these problems.

2.1 PROBLEM DEFINITION

Manual tracking of job applications leads to:

- **loss of application details**
- **confusion in interview schedules**
- **difficulty tracking status**

This system helps manage applications digitally.

2.2 PROJECT OVERVIEW

The project provides a simple platform for students to manage job application details.

Students can:

Enter subject marks and credits

- **Calculate semester GPA**
- **Calculate overall CGPA**

2.3 SCOPE AND LIMITATIONS

SCOPE

Useful for students to monitor academic performance

- **Helps calculate GPA and CGPA accurately**
- **Simple and user-friendly interfacE**

LIMITATIONS

No database storage

- **Data entered temporarily during execution**
- **Console-based application**

SYSTEM DESIGN

System design explains structure and working of the project.

The system is divided into modules.

3.1 MODULES

1. Add Application Module

This module allows students to enter:

Subject name

- Subject marks
- Subject credits

---

2. View Application Module

This module displays all saved marks applications.

---

3. Update Status Module

This module allows updating job status like:

- **Applied**
- **Interview Scheduled**
- **Selected**
- **Rejected**

---

4. Exit Module

This module stops program execution.

METHODOLOGY

Steps followed during development:

- **Problem analysis**
- **System design**
- **Program development**
- **Testing**
- **Documentation**

4.1 PROBLEM ANALYSIS

Students applying for multiple companies cannot track application details properly.

This system solves that issue.

4.2 SYSTEM DESIGN

The system is designed as a menu-driven Java console application.

Modules included:

- **Add application**
- **View application**
- **Update status**

4.3 PROGRAM DEVELOPMENT

This system is developed using:

- **Classes and objects**
- **ArrayList**
- **Scanner class**
- **Loops**
- **Conditional statements**

4.4 TESTING AND VALIDATION

Testing ensures program works correctly.

Test cases:

Test Case	Expected Result
Add cgpa-gpa	Stored successfully
View cgpa-gpa	Display correctly
Update status	Updated successfully

4.5 RESULT AND DOCUMENTATION

The program successfully:

- **stores cgpa-gpa details**
- **displays cgpa-gpa list**
- **updates cgpa-gpa status**

4.6 ALGORITHM

Algorithm steps:

Start program
Display menu
Enter choice

Enter semester details
→ Enter subject details
→ Enter total and obtained marks

Calculate GPA
→ Display semester GPA

Calculate CGPA
→ Display overall percentage and CGPA

Repeat until exit

Stop program

4.7 FLOWCHART

Flow:

Start
↓
Display menu
↓
Add / View / Update
↓
Repeat
↓
END

SOURCE CODE

```
import java.util.Scanner;


public class Main {
```

```
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);

System.out.println("=== GPA & CGPA CALCULATOR ===");

System.out.print("Enter number of semesters: ");
int semesters = sc.nextInt();
sc.nextLine();

double overallPercentage = 0;

double totalObtained = 0;
double totalMaximum = 0;

for (int i = 1; i <= semesters; i++) {
System.out.println("\n--- Semester " + i + " ---");

System.out.print("Enter number of subjects: ");
int subjects = sc.nextInt();
sc.nextLine();

double semesterObtained = 0;
double semesterMaximum = 0;

for (int j = 1; j <= subjects; j++) {
System.out.println("\nSubject " + j);
```

```
System.out.print("Enter subject name: ");
String name = sc.nextLine();

System.out.print("Enter total marks: ");
double maximum = sc.nextDouble();

System.out.print("Enter marks obtained: ");
double obtained = sc.nextDouble();
sc.nextLine();

if (maximum <= 0 || obtained < 0 || obtained > maximum) {
System.out.println("Invalid marks. Please restart.");
sc.close();
return;
}

semesterMaximum += maximum;
semesterObtained += obtained;

totalMaximum += maximum;
totalObtained += obtained;
}

double gpa = (semesterObtained / semesterMaximum) * 10;
```

```
System.out.printf(
"\nSemester %d GPA: %.2f / 10%n", i, gpa
);
System.out.printf(
"Semester %d Percentage: %.2f%%%n",
i, (semesterObtained / semesterMaximum) * 100
);
}

double cgpa = (totalObtained / totalMaximum) * 10;

System.out.println("\n=== FINAL RESULT ===");
System.out.printf("Total Marks: %.2f%n", totalMaximum);
System.out.printf("Marks Obtained: %.2f%n", totalObtained);
System.out.printf("Overall Percentage: %.2f%%%n",
(totalObtained / totalMaximum) * 100);
System.out.printf("Overall CGPA: %.2f / 10%n", cgpa);

sc.close();
}
}
```

INPUT AND OUTPUT

```
import java.util.Scanner;

public class Main {
public static void main(String[] args) {
 Scanner sc = new Scanner(System.in);

System.out.println("=== GPA & CGPA CALCULATOR ===");

System.out.print("Enter number of semesters: ");
int semesters = sc.nextInt();
sc.nextLine();

double overallPercentage = 0;

double totalObtained = 0;
double totalMaximum = 0;

for (int i = 1; i <= semesters; i++) {
System.out.println("\n--- Semester " + i + " ---");

System.out.print("Enter number of subjects: ");
```

```
int subjects = sc.nextInt();
sc.nextLine();

double semesterObtained = 0;
double semesterMaximum = 0;

for (int j = 1; j <= subjects; j++) {
System.out.println("\nSubject " + j);

System.out.print("Enter subject name: ");
String name = sc.nextLine();

System.out.print("Enter total marks: ");
double maximum = sc.nextDouble();

System.out.print("Enter marks obtained: ");
double obtained = sc.nextDouble();
sc.nextLine();

if (maximum <= 0 || obtained < 0 || obtained > maximum) {
System.out.println("Invalid marks. Please restart.");
sc.close();
return;
}

semesterMaximum += maximum;
```

```
semesterObtained += obtained;

totalMaximum += maximum;
totalObtained += obtained;
}

double gpa = (semesterObtained / semesterMaximum) * 10;

System.out.printf(
"\nSemester %d GPA: %.2f / 10%n", i, gpa
);
System.out.printf(
"Semester %d Percentage: %.2f%%%n",
i, (semesterObtained / semesterMaximum) * 100
);
}

double cgpa = (totalObtained / totalMaximum) * 10;

System.out.println("\n=== FINAL RESULT ===");
System.out.printf("Total Marks: %.2f%n", totalMaximum);
System.out.printf("Marks Obtained: %.2f%n", totalObtained);
System.out.printf("Overall Percentage: %.2f%%%n",
(totalObtained / totalMaximum) * 100);
System.out.printf("Overall CGPA: %.2f / 10%n", cgpa);
```

```
sc.close();
    }
}
```

INPUT AND OUTPUT

=== GPA & CGPA CALCULATOR ===

Enter number of semesters: 2

--- Semester 1 ---

Enter number of subjects: 2

Subject 1

Enter subject name: Java

Enter total marks: 100

Enter marks obtained: 85

Subject 2

Enter subject name: Python

Enter total marks: 100

Enter marks obtained: 90

Semester 1 GPA: 8.75 / 10

Semester 1 Percentage: 87.50%

--- Semester 2 ---

Enter number of subjects: 2

Subject 1

Enter subject name: DBMS

Enter total marks: 100

Enter marks obtained: 80

Subject 2

Enter subject name: Web Technology

Enter total marks: 100

Enter marks obtained: 95

Semester 2 GPA: 8.75 / 10

Semester 2 Percentage: 87.50%

=== FINAL RESULT ===

Total Marks: 400.00

Marks Obtained: 350.00

Overall Percentage: 87.50%

Overall CGPA: 8.75 / 10

DISCUSSION AND CONCLUSION

Discussion

The system helps students manage cgpa-gpa calculator applications efficiently.

It reduces confusion and improves organization.

Conclusion

The GPA and CGPA Calculator is a simple Java-based console application that helps students calculate their semester-wise GPA and overall CGPA effectively.

It demonstrates the practical use of Core Java programming concepts to solve real-world academic calculation problems.

REFERENCES

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Programming with Java – E. Balagurusamy

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www.geeksforgeeks.org
www.w3schools.com/java